SYSTEMS BIOLOGY IN REPRODUCTIVE MEDICINE
2024, VOL. 70, NO. 1, 38–51
https://doi.org/10.1080/19396368.2024.2306389

Taylor & Francis
Taylor & Francis Group

REVIEW

Check for updates

Preimplantation genetic testing as a preventive strategy for the transmission of mitochondrial DNA disorders

Xinpeng Lan[a], Wu Liji Ao[b], and Ji Li[c]

[a]College of Basic Medical Sciences, Heilongjiang University of Chinese Medicine, Harbin, Heilongjiang, China; [b]College of Mongolian Medicine and Pharmacy, Inner Mongolia University for Nationalities, Tongliao, Inner Mongolia, China; [c]College of Basic Medical Sciences, Heilongjiang University of Chinese Medicine, Harbin, China

ABSTRACT

Mitochondrial diseases are distinct types of metabolic and/or neurologic abnormalities that occur as a consequence of dysfunction in oxidative phosphorylation, affecting several systems in the body. There is no effective treatment modality for mitochondrial disorders so far, emphasizing the clinical significance of preventing the inheritance of these disorders. Various reproductive options are available to reduce the probability of inheriting mitochondrial disorders, including *in vitro* fertilization (IVF) using donated oocytes, preimplantation genetic testing (PGT), and prenatal diagnosis (PND), among which PGT not only makes it possible for families to have genetically-owned children but also PGT has the advantage that couples do not have to decide to terminate the pregnancy if a mutation is detected in the fetus. PGT for mitochondrial diseases originating from nuclear DNA includes analyzing the nuclear genome for the presence or absence of corresponding mutations. However, PGT for mitochondrial disorders arising from mutations in mitochondrial DNA (mtDNA) is more intricate, due to the specific characteristics of mtDNA such as multicopy nature, heteroplasmy phenomenon, and exclusive maternal inheritance. Therefore, the present review aims to discuss the utility and challenges of PGT as a preventive approach to inherited mitochondrial diseases caused by mtDNA mutations.

Abbreviations: CNS: Central nervous system; ddPCR: Droplet digital polymerase chain reaction; IVF: In vitro fertilization; KSS: Kearns-Sayre syndrome; LHON: Leber's hereditary optic neuropathy; MELAS: Lactic acidosis, stroke-like episodes; MERRF: Myoclonic epilepsy, with ragged red fibers; mtDNA: Mitochondrial DNA; NARP: Neurogenic weakness with ataxia and retinitis pigmentosa; nDNA: Nuclear DNA; NGS: Next-generation sequencing; PB: Polar body; PGC: Primordial germ cell; PGT: Preimplantation genetic testing; PND: Prenatal diagnosis; scATAC-Seq: Single-cell transposase-accessible chromatin with sequencing; scDNA-Seq: Single-cell DNA sequencing; scRNA-Seq: Single-cell RNA sequencing; TE: Trophectoderm

ARTICLE HISTORY
Received 30 August 2023
Revised 13 December 2023
Accepted 7 January 2024

KEYWORDS
Preimplantation diagnosis; mtDNA diseases; Heteroplasmic mutations; Maternal inheritance; Mitochondria

Introduction

Mitochondria are major components of eukaryotic cells mainly characterized as energy-producing units within the cell through oxidative phosphorylation. They are involved in a variety of cellular procedures, including the provision of ATP, the regulation of intracellular Ca^{2+} homeostasis, and the formation of reactive oxygen species (Kann and Kovács 2007), which in turn are essential for the conduction of electrical signals along neurons (Zhu et al. 2021), the contractile activity of cardiomyocytes (Wang et al. 2018), endothelial function (Chen et al. 2021). Mitochondrial diseases are a distinct type of metabolic and/or neurologic abnormalities that result from dysfunctional oxidative phosphorylation and influence several organs, especially those that have a high energy demand like the brain, the heart, skeletal muscles, and the liver (McFarland et al. 2010). They usually involve numerous systems and show a wide range of rapidly growing clinical manifestations, leading to early morbidity and mortality (Gorman et al. 2016). Mitochondrial diseases are also heterogeneous in terms of genotype and phenotype. Kearns-Sayre syndrome (KSS), Leigh syndrome, Leber's hereditary optic neuropathy (LHON), myoclonic epilepsy, lactic acidosis, stroke-like episodes (MELAS), myoclonic epilepsy, with ragged red fibers (MERRF), neurogenic weakness with ataxia and retinitis pigmentosa (NARP), and macular dystrophy are well-established clinical syndromes arising

CONTACT Ji Li lxp18645020081@163.com College of Basic Medical Sciences, Heilongjiang University of Chinese Medicine, Harbin 150036, China
© 2024 Informa UK Limited, trading as Taylor & Francis Group

out of mitochondrial disorders (Finsterer 2011; Xiang et al. 2021). Nevertheless, clinical symptoms are frequently different between affected individuals and overlap with other disorders, conflicting with discrete clinical conditions (Craven et al. 2017).

Mitochondrial diseases show genetic heterogeneity and can result from the mutations of both nuclear DNA (nDNA) and mitochondrial DNA (mtDNA). With respect to epidemiology and prevalence, childhood-onset mitochondrial diseases were estimated to occur in 5 to 15 per 100,000 children, while this statistic is estimated to be 12.5 per 100,000 individuals in the case of adulthood-onset mitochondrial diseases (Gorman et al. 2015, 2016). Despite the increase in our knowledge about the genetic aspects of mitochondrial disorders and considerable progress in the scope of mitochondrial medicine such as novel therapeutic modalities, there is no efficacious therapy for mitochondrial diseases so far, and the treatment is confined to the alleviation of symptoms (Nightingale et al. 2016), highlighting the necessity of preventing the transmission of these diseases.

There are different reproductive choices for women who carry mitochondrial disease-related mutations to decrease the chance of disease transmission, including *in vitro* fertilization (IVF) using donated oocytes, mitochondrial replacement therapy (also known as mitochondrial donation), preimplantation genetic testing (PGT), and prenatal diagnosis (PND) (Craven et al. 2017). PGT and PND provide the possibility that patients and/or carriers of mitochondrial disorders have healthy and genetically-owned babies. PND has the disadvantage that couples may have to terminate the pregnancy due to the identification of an affected fetus. Conversely, PGT offers the possibility to transfer embryos diagnosed to lack pathogenic mutation loads and thus significantly reduces the probability of developing mitochondrial diseases in the fetus (Smeets et al. 2015).

PGT for nDNA-related mitochondrial disorders requires evaluating the existence or lack of responsible mutation. However, PGT for mitochondrial diseases caused by mtDNA mutations necessitates quantifying the abundance of wild-type and mutant mtDNAs, as the cells of most individuals with mtDNA mutation accommodate different proportions of wild-type and mutant mtDNA molecules, a phenomenon called heteroplasmy. Disorders will manifest if the proportion of mutant mtDNA molecules passes certain thresholds, which are distinct for each mutation (Stewart and Chinnery 2015). Accordingly, in addition to embryos containing wild-type mtDNAs, those that have lower frequencies of mutation and thus are improbable to result in disease can qualify for uterine transfer (Poulton and Bredenoord 2010). Sibling embryos can be significantly different in terms of mutation load probably due to the bottleneck effect during oogenesis (Poulton et al. 1998; Stewart and Chinnery 2015). Therefore, PGT can be beneficial for women suffering from mtDNA disorders as well as those who asymptomatically harbor pathogenic mtDNA mutations, as affected offspring may be born from both groups (Dahl et al. 2000). Hence, the present review article aimed to discuss the applicability of PGT as a preventive measure to heritable mitochondrial diseases caused by mtDNA mutations.

Mitochondrial DNA

Mitochondria are special organelles since they contain their own genetic material, i.e., mtDNA. There are multiple specific properties that differentiate mtDNA from the nuclear genome. Particularly, mtDNA is a compressed double-helix circular molecule with an approximate length of 16.6 Kbp, which has unique genetic codes. A total of 37 genes exist in mtDNA, among which 13 genes encode oxidative phosphorylation constituents and the rest are responsible for encoding mitochondrial translation machinery, including 22 tRNA and 2 rRNA genes (Calvo et al. 2016). The lack of intron sequences along with the existence of overlapping regions and incomplete termination codons are considered other characteristics of mtDNA (Smeets et al. 2015). Besides proteins encoded by the mitochondrial genome, about ~1145 other proteins, which are encoded by the nuclear genome, are required for mitochondrial activity, indicating that mitochondrial activity is genetically regulated by both mtDNA and nDNA (Calvo et al. 2016). MtDNA has another key property that plays a key role in the occurrence of mtDNA disorders, and this includes the multicopy essence of mtDNA. Accordingly, there is a varying range of mtDNA molecules in cells depending on the type of tissue that cells belong to. Human gametes are typical examples of the fact that numerous mtDNA copies are present in human cells, ranging from ~100 mtDNA molecules in sperm (Wu H et al. 2019) and ~100,000–600,000 mtDNA copies in mature oocytes (Reynier et al. 2001). Importantly, these mtDNA molecules can genotypically be identical or diverse, the situations that are called homoplasmy and heteroplasmy, respectively. These conditions are of clinical importance when mtDNA contains pathogenic mutations (Smeets et al. 2015; Craven et al.

2017). Homoplasmic mutations (such as 3460 G→A, 11778 G→A, or 14484 T→C, which affect different subunits of complex I and lead to LHON) involve all existing mtDNA molecules, being inherited by the mother's offspring without exception, whereas heteroplasmic ones are present in a specific proportion of mtDNA pool (Taylor and Turnbull 2005). In the case of heteroplasmy, the incidence of disorders depends on the ratio of mutant mtDNA to its wild-type counterpart, and the severity of manifestations is directly linked to mutation levels. This phenomenon is recognized as the threshold effect, in which clinical symptoms appear if the mutation load passes a crucial threshold. In general, mutation load between 60% and 80% is speculated as threshold values, however, it can change with the type of mtDNA mutation and is yet to be determined in the case of some disorders (Smeets et al. 2015; Craven et al. 2017).

Another unique characteristic of mtDNA is that it is exclusively inherited from the mother (Pyle et al. 2015), presumably owing to the programmed degradation of sperm mtDNA molecules within fertilized oocytes (Song et al. 2016; Zhou et al. 2016). This substantially increases the risk of inheriting mutation and developing mitochondrial disease by the offspring of women with pathogenic mutant mtDNA. Furthermore, the predictability of this risk is notably low due to another specific feature of mtDNA inheritance. Accordingly, the population of mtDNA molecules undergoes a reduction during the formation of the female germline, a phenomenon termed mitochondrial genetic bottleneck (Cree et al. 2008; Wai et al. 2008; Neupane et al. 2015; Zhang H et al. 2018), giving rise to the distribution of a restricted number of mtDNA entities in primordial germ cells (PGCs). The genetic bottleneck effect can bring about extensive alterations in mtDNA heteroplasmy in both the current and the next generation. For this account, genetic counseling for female carriers of pathogenic mutations in mtDNA is difficult since even asymptomatic women harboring declined levels of mutation may generate oocytes with elevated levels of mutation. Moreover, the levels of mtDNA heteroplasmy obviously differ among the offspring of women carrying disease-causing mtDNA mutations, which makes genetic counseling for these individuals more challenging. Indeed, the difference in the transmission of pathogenic mutations and the lack of unanimous threshold limits for mutation loads can lead to the incidence of mitochondrial disorders and organ failure (Craven et al. 2017).

Mitochondrial disorders

There are a large number of mitochondrial diseases resulting from genetic errors in mtDNA. In the following section, mitochondrial diseases that can have an mtDNA origin and were subjected to PGT cycles in previous studies are briefly introduced.

Leigh syndrome

Leigh syndrome is typically manifested in a range of 3 months to 2 years old of infants, representing the most prevalent mtDNA disease which has onset in childhood. Although the majority of individuals affected by Leigh syndrome manifest the disease during infancy, there are cases in which the disease shows a rapid progression immediately after birth or has an onset in adolescence or the beginning of adulthood (Gorman et al. 2016). Leigh syndrome is principally defined as a neurodegenerative disorder with different clinical presentations owing to the dysfunction of mitochondria arising from inherited genetic errors that result in bilateral damage to the central nervous system (CNS). Clinically, Leigh syndrome is recognized by neurodevelopmental deterioration, appearing with impairments in brainstem function like anomalies in tone, power, reflexes, ataxia, dysphagia, and seizures (Bakare et al. 2021). Although they can vary among affected cases, clinical symptoms indicate the brain's regions, including the brainstem, cerebellum, basal ganglia, oculomotor, and cranial nerves, that are implicated in Leigh syndrome's pathology (Ruhoy and Saneto 2014). Genetic defects in at least 75 distinct genes located in either mtDNA or nDNA are responsible for the occurrence of Leigh syndrome (Lake et al. 2016). Specifically, mtDNA mutations in the *MT-ATP6*, *MT-TL1*, *MT-TK*, *MT-TY*, *MT-TV*, *MT-ND1*, *MT-ND3*, *MT-ND4*, *MT-ND5*, *MT-ND6*, or *MT-CO3* genes engender Leigh syndrome (Finsterer 2011).

NARP (neurogenic muscle weakness, ataxia, and retinitis pigmentosa)

NARP is a developing neurodegenerative disease, and its onset usually occurs in infancy, but it can remain asymptomatic or constant until adulthood (Gorman et al. 2016). Dominant clinical manifestations of NARP are as follows: muscular deficiency because of polyneuropathies in sensory and motor neurons, ataxia because of cerebellar atrophy, and the loss of vision because of either optic atrophy, salt and pepper retinopathy, bull's eye maculopathy, or retinitis pigmentosa (Holt et al. 1990). In addition, NARP may be

associated with some inconsistent presentations, including a decrease in stature, paralysis of the eye muscles, learning disabilities, dementia, sleep apnea, seizures, or cardiac arrhythmias (Santorelli et al. 1997). Despite a slow progression of the process, deterioration intervals may come to the fore owing to infections. NARP has a maternal pattern of heredity and is predominantly caused by the heteroplasmic transversion or transition of the *MT-ATP6* gene in mtDNA, i.e., m.8993T→G and m.8993T→C, respectively (Finsterer 2011). Both of these mutant variants display the closest genotype-phenotype relationship in comparison to other mtDNA mutations, indicating that mtDNA mutation loads are directly correlated with the severity of the disease (Tatuch et al. 1992; White et al. 1999). NARP is intimately linked to Leigh syndrome and together they create a clinical spectrum. Generally, subjects who contain less than 70% of the m.8993T→G variant do not show any symptoms, heteroplasmy levels between 70% and 90% clinically appear as NARP phenotype, while individuals harboring over 90% mutation loads exhibit clinical characteristics representing Leigh syndrome (White et al. 1999). It is of interest to note that, the transition m.9185T→C in the *MT-ATP6* gene can also give rise to the NARP phenotype (Childs et al. 2007).

MELAS (mitochondrial encephalomyopathy, lactacidosis, and stroke-like-episodes)

MELAS is a disorder that follows a maternal inheritance pattern, affects multiple systems in the body, and with respect to clinical manifestation, is accompanied by stroke-like-episodes, tonic-clonic seizures, migraine-like headache, confusional state, dementia, psychosis, deafness, short stature, recurrent vomiting, or polyneuropathies. Heteroplasmic m.3243A→G mutation in the mitochondrial *MT-TL1* gene is the most prevalent cause of MELAS, accounting for approximately 80% of patients. In order of frequency, MELAS also occurs as a consequence of m.3271T→C and m.3252A→G mutations in the mitochondrial *MT-TL1* gene (Finsterer 2011).

PGT for mtDNA disorders

PGT, which was introduced more than thirty years ago (Handyside et al. 1990), is a diagnostic tool that provides an opportunity to select and transfer mutation-free embryos obtained from IVF, leading to the elimination or reduction of transmitting genetic disorders to offspring (Craven et al. 2017). With the help of PGT, families harboring genetic abnormalities can avoid being confronted with a decision regarding pregnancy termination if the fetus is affected (Smeets et al. 2015). The process of PGT comprises the removal of one or two blastomeres from an early embryo followed by genetic analyses of them, and ultimately opting for embryos without mutation or with low levels of mutation for uterine transfer (Hellebrekers et al. 2012). From the technical point of view, PGT for mtDNA disorders is simpler and is accompanied by fewer faults such as PCR failure or allelic dropout as compared to nDNA-related diseases, since there are a large number of mtDNA molecules (about 75,000 copy number) in each blastomere, only two copy numbers of a nuclear gene exist (Smeets et al. 2015). However, PGT will be efficacious and reliable if heteroplasmy levels of blastomeres reflect those of the whole embryo, which in turn needs a fairly equal distribution of mtDNA mutation throughout early embryogenesis (Craven et al. 2017). The following sections are devoted to delineating evidence on the dynamics and mechanisms of mtDNA segregation during early embryo development in animal models and humans as well as address the utility of the PGT approach in clinical settings.

Studies on animal models

Dean et al. (2003) showed that mtDNA variants were uniformly segregated between polar bodies (PBs) and mature oocytes, as well as among blastomeres of cleaved embryos in heteroplasmic mouse models. Importantly, the analysis of blastomeres yielded a more reliable assessment regarding mtDNA heteroplasmy in comparison with the first PBs of oocytes which failed to be fertilized, suggesting the potential feasibility of PGT for evaluating mtDNA segregation. Consistently, Neupane et al. (2014) in their study on heteroplasmic mouse models found robust correlations among single blastomeres of 2-, 4-, and 8-cell embryos, as well as between trophectoderm (TE) and the entire blastocyst in terms of mtDNA heteroplasmy. Furthermore, the mtDNA diversity of individual blastomeres from 8-cell embryos was closely correlated with that of relevant TE cells and the entire blastocysts, implying that PGT based on blastomere or TE can be a reliable tool for determining mtDNA variant distribution. However, the authors found poor correlations between PBs and corresponding MII oocytes, zygotes, blastomeres, TEs, and blastocysts, questioning the reliability of PGT based on the analysis of first and second PBs. In sharp contrast, Lee

et al. (2012) reported a significant difference in mtDNA heteroplasmy level between blastomeres of 8-cell embryos in heteroplasmic monkey models. They additionally noticed a considerable transition from a heteroplasmic situation to a homoplasmic situation in fetuses of relevant embryos, indicating the instability of mtDNA variation during embryo-fetal growth and development. Nevertheless, it was argued that the abnormal segregation of mtDNA observed in the previous study might stem from the specific nature of the artificially-generated models used (Steffann et al. 2014). This indicates the fact that there might be a specific segregation pattern depending on the type of mtDNA variant, which necessitates interpreting the findings obtained from animal models with caution.

Studies on humans

As mentioned previously, in order to prevent the expression of mtDNA diseases, it is not necessarily required to transfer only mutation-free embryos, but embryos containing lower mutation loads than the clinical threshold are transferable (Poulton and Bredenoord 2010). According to a meta-analysis, which assessed 159 distinct heteroplasmic mtDNA single-nucleotide mutations existed in 327 separate families, regardless of the type of mutation, muscular heteroplasmy levels of 18% or lower are associated with 95% or higher probability of being unaffected (Hellebrekers et al. 2012). Likewise, a recent work estimated thresholds of 29.86% and 16.15% for the m.8993T→G and m.8344A→G mutations, respectively, below which heteroplasmy does not lead to clinical manifestation (Ji et al. 2023). In addition, several lines of evidence demonstrated the stability of mtDNA heteroplasmy during human embryofetal development, regardless of the type of mutation (Steffann et al. 2007; Monnot et al. 2011; Steffann et al. 2021), supporting the validity of PGT for estimating mtDNA heteroplasmy in resultant offspring. It is worth mentioning that the existence of some levels of mtDNA heteroplasmy in transferred embryos exposes the next female generation to the risk of transmitting mtDNA mutation. For this reason, male embryos are usually identified by performing a gender determination test during PGT and selected for transfer in order to completely rule out the possibility of maternal transmission of mtDNA mutations to the next generations.

Due to potential differences in inheritance properties between various types of mtDNA mutations and illustrating the correlation between PGT results and newborn's genotype, other clinical evidence on inter-blastomere heteroplasmy are described separately according to the type of mutation in the following (Table 1).

The m.8993T→G mutation

The m.8993T→G mutation is one of the most prevalent pathogenic mtDNA variants that result in Leigh syndrome or NARP based on the degree of heteroplasmy. The first attempt at performing PGT was made by Steffann et al. (2006) on a mother who harbored 18% heteroplasmy in blood for the m.8993T→G mutation and had a child with NARP. Accordingly, mutant mtDNA displayed a bipolar segregation pattern among IVF-generated embryos, as one of the embryos was homoplasmic for mutant mtDNA, whereas the remaining two embryos were homoplasmic for wild-type mtDNA based on the analysis of two blastomeres from cleaved embryos. Uterine transfer of the latter embryos led to a singleton gestation followed by the birth of a male infant. Neither amniotic fluid nor cord blood samples exhibited evidence of the m.8993T→G mutation, confirming the results obtained from PGT. The baby was followed up until the age of seven months and was in good health during that time.

Similar to the findings of the previous study, data on PGT cycles presented by Sallevelt et al. (2013) on a woman with 5% heteroplasmy in urine for the m.8993T > G mutation showed that this mutation followed a skewed pattern of distribution in oocytes and embryos, as only one of five oocytes and three of twenty embryos contained the mutation with the mean load of 16% and 30–100%, respectively. Moreover, the heteroplasmy level was comparable between single blastomeres of each embryo (interblastomere disparity <10%). Three embryos were transferred to the patient's uterus, resulting in one pregnancy with the delivery of a healthy boy. No mutation was found in the analysis of cord blood, which accorded with the results of PGT.

In a study on 13 normal individuals, Tajima et al. (Tajima et al. 2007) found that the m.8993T→G mutation was present in 5 out of 20 embryos, representing a transmission rate of 25%, with heteroplasmy levels ranging from 4 to 22%. In addition, while mutation load differed among embryos of a particular person, it showed a limited range of variation (2–11%) among cleavage-stage blastomeres of a given embryo, which supports the efficacy of PGT for m.8993T→G mutation.

Table 1. **Clinical PGT cycles carried out for mtDNA disorders.**

Reference	mtDNA mutation	Disorder	Number of patients	Mutation load of embryos	Mutation load difference	Mutation load of transferred embryo(s)	Mutation load of fetal/ neonatal tissues
(Steffann et al. 2006)	m.8993T→G	NARP/Leigh syndrome	1	0–100%	6% between two cleavage-stage blastomeres and the entire embryo	0%	0% in cord blood
(Sallevelt et al. 2013)	m.8993T→G	NARP/Leigh syndrome	1	30–100%	<10% between cleavage-stage blastomeres	0%	0% in cord blood
(Tajima et al. 2007)	m.8993T→G	NARP/Leigh syndrome	13	4–22%	2–11% between cleavage-stage blastomeres	NA	NA
(Spath et al. 2021b)	m.8993T→G	NARP/Leigh syndrome	2	0–88.7%	NA	0%	0% (not specified in which tissue)
(Sallevelt et al. 2013)	m.3243A→G	MELAS	3	1–70%	<10% (80% of embryos) and >15% (18% of embryos) between cleavage-stage blastomeres	NA	NA
(Monnot et al. 2011)	m.3243A→G	MELAS	2	5–77%	0–6% between cleavage-stage blastomeres 0–2% between cleavage-stage blastomeres and TE cells <7% between cleavage-stage blastomeres and the entire blastocyst	NA	NA
(Treff et al. 2012)	m.3243A→G	MELAS	1	7–90%	0.5–2.9% between cleavage-stage blastomeres2.1–5.0% between TE cells and inner cell mass	12%	0–15% in buccal cells, blood, and urine
(Heindryckx et al. 2014)	m.3243A→G	MELAS	1	0–65%	0–7% between TE cells	0%	0% in saliva, blood cells, and urine
(Spath et al. 2021a)	m.3243A→G	MELAS	1	3.1–29.2%	0–3.2% between halves of cleavage-stage embryos	NA	NA
(Sallevelt SC et al. 2017)	m.14487T→C	Leigh syndrome	1	6–100%	0–10% between cleavage-stage blastomeres	<18%	5% mutation load in cord blood
(Spath et al. 2021a)	m.10191T→G	Leigh syndrome	1	0%	0%	0%	0% in chorionic villi
(Ji et al. 2021)	m.3697G→A	Leigh syndrome	1	15.2%–100%	0–10.3% between cleavage-stage blastomeres	53.9%	47.7–49.7% in fetal tissues and amniotic fluid cells

Recently, Spath et al. (2021) performed concurrent PGT for mtDNA disorder and aneuploidy in two carriers of the m.8993T→G mutation with 56% and undetectable heteroplasmy in blood. Accordingly, patient A yielded four oocytes, of which three were fertilized, leading to two arrested embryos and one blastocyst. While the blastocyst and one of the two arrested embryos were found to be mutation-free, the unfertilized oocyte and the other arrested embryo showed elevated mutation loads (91.2% and 88.7%, respectively). Despite homoplasmy of blastocyst for wild-type mtDNA, it was denied to be transferred due to aneuploidy. In patient B, however, 18 oocytes were retrieved, resulting in 17 zygotes and 14 blastocysts. All samples were devoid of the m.8993T→G mutation upon genetic analysis. A healthy son was born as a result of transferring one embryo, as evidenced by the lack of mutation in none of the neonate's specimens.

Overall, it can be concluded that the m.8993T→G mutation is segregated among oocytes and corresponding zygotes in an almost nothing-or-all fashion, exhibits an ignorable level of variation among blastomeres of a given embryo, and has an extremely strong embryo-newborn correlation in terms of mutation load. Therefore, PGT seems to be an efficacious option for preventing the expression of Leigh syndrome or NARP in offspring of m.8993T→G carriers.

The m.3243A→G mutation

The m.3243A→G transition is a mutation that occurs in the *MT-TL1* gene and thereby affects the production of mitochondrial tRNA for leucine, accounting for the primary cause of a prevalent and serious heteroplasmic mtDNA disease called MELAS. Results of PGT in three women with 13%, 25%, and 27% heteroplasmy in blood for the m.3243A→G mutation showed a 100% transmission rate with a heteroplasmy ranges of 37–67%, 0–66%, and 1–70% in oocytes, zygotes, and cleavage-stage embryos, respectively. In addition, while the mutation load of sister blastomeres was relatively concordant in around 80% of embryos (interblastomere variation <10%), it varied significantly in 18% of embryos, which exhibited >15% difference in heteroplasmy level. Eventually, Three embryos from two patients were transferred, none of which led to pregnancy achievement (Sallevelt et al. 2013). Consistently, the analysis of 73 embryos derived from 9 women harboring the m.3243A→G mutation demonstrated that among 294 individual blastomeres, only one displayed a false-negative result according to the threshold point of 15%, whereas no blastomere yielded a false-positive result, indicating

the relative stability of mutation load in the majority of cleaved embryos, as well as the adequate diagnostic accuracy of PGT based on one blastomere biopsy (Sallevelt et al. 2017).

Similarly, Monnot et al. (2011) in their study on two heteroplasmic female carriers of the m.3243A→G mutation with heteroplasmy of 65% and 30% in urine found that the variant was present in 84% of embryos, which all contained different loads of mutation ranging from 5% to 77%. The maximum variation in heteroplasmy of blastomeres biopsied from a given cleaved embryo was 6%, indicating that there was no significant interblastomere difference. In addition, the variation of mutation load was 2% between cleavage-stage blastomeres and TE cells, and < 7% between blastomeres and the entire blastocysts for the majority of embryos, suggesting that single blastomeres of cleaved embryos could reflect the heteroplasmy of the resultant entire blastocysts. Mutation load also remained almost constant at different time points of pregnancy and was equal among various fetal tissues.

In keeping with these findings, another study on the application of PGT in a woman carrying 35% heteroplasmy for the m.3243A→G mutation found that there were wide ranges of mutation loads in oocytes (9–90%) and cleavage-stage embryos (7–86%), and blastocysts (12–90%). Interblastomere difference in heteroplasmy level was in the range of 0.5% to 2.9%, suggesting that a single blastomere could reflect the mutation load of the whole cleaved embryo. In addition, heteroplasmy differences ranged from 2.1% and 5.0%, and mutation load was strongly correlated between TE cells and inner cell mass, indicating the reliability of TE cells as representative of the entire blastocyst genotype. Transfer of a male blastocyst containing 12% heteroplasmy based on TE analysis led to the birth of a son harboring a range of 0 to 15% mutation load depending on the type of tissue (Treff et al. 2012). Nevertheless, controversy arose over the level of heteroplasmy in the child later, he was found to have a heteroplasmy range of 42%–52% at age 6 weeks and 18 months following the reassessment in another center, respectively (Mitalipov et al. 2014).

Heindryckx et al. (2014) also studied the utility of PGT using TE biopsy for determining heteroplasmy in the offspring of a carrier woman with 16% heteroplasmy in blood and found a range of 0% to 65% mutation loads in blastocysts. Heteroplasmy variation among sister TE cells ranged from 0% to 7%, with an outlier that was equal to 22.7%. There was also a robust correlation between TE cells of a given embryo in terms of mutation load. Interestingly, the heteroplasmy level of nucleus-free cytoplasmic fragments was intimately correlated with and consequently was representative of that of TE cells. Two blastocysts with wild-type homoplasmic genotypes were transferred, and as a consequence, a healthy boy was born. The saliva, blood cells, and urine of the baby had a 0% mutation load.

In contrast to the previous findings, the transmission rate of the m.3243A→G mutation from a carrier mother with 28% heteroplasmy in muscle to offspring was found to be one-third in a recent study by Spath et al. (2021), in which the range of blastocyst heteroplasmy was 3.1–29.2%. However, consistent with the literature, the authors observed a minor difference in mutation load (ranging from 0 to 3.2%) between halves of arrested cleavage-stage embryos. Two mutation-free euploid embryos were transferred, but miscarriage occurred at gestational week 8 in spite of successful implantation and the detection of the fetal heartbeat.

Altogether, it can be concluded that the m.3243A→G mutation has an extremely high transmission rate, affecting the majority of carriers' embryos but with a wide range of mutation loads. Small differences in heteroplasmy level exist between cleavage-stage blastomeres, between cleavage-stage blastomeres and TE cells, between TE cells, and between TE cells and inner cell mass, indicating that an individual blastomere or TE cell represents the mutation load of the entire embryo. Moreover, the results of PGT show thorough compatibility with the genotype of newborns, implying that PGT can be helpful in preventing the transmission of the m.3243A→G mutation and thus the occurrence of MELAS in the offspring of affected women.

The m.14487T→C mutation

The m.14487T→C mutation occurs in the *MT-ND6* gene, a gene that encodes NADH dehydrogenase subunit 6, which is a constituent of complex I, and gives rise to Leigh syndrome. Sallevelt et al. (2017) presented the first and only report on the PGT procedure for this mutation in a female carrier with 32%, 55%, and 65% heteroplasmy in lymphocytes, muscle, and urine, respectively. According to the authors, in the first cycle, the mutation was present in 100% of cleaved embryos with high levels of mutation loads ranging from 50% to 98%, which makes embryos not transferable. Consistently, all cleavage-stage embryos contained varied levels (6–100%) of heteroplasmy in the second cycle. Collectively, maximal variation in mutation load ranged from 0 to 5% in the majority of

embryos and was <10% in the vast majority of embryos. A singleton pregnancy was achieved following the transfer of two embryos harboring less than 18% heteroplasmy, leading to the birth of a healthy male infant. About 5% mutation load was detected in the baby's cord blood, which was similar to that of one of the transferred embryos.

Taken together, it can be inferred from these findings that the m.14487T→C mutation has a high transmission potential, but it does not negate the applicability of PGT for this mutation due to the concordance of heteroplasmy between blastomeres and neonatal tissues.

The m.10191T→G mutation

The m.10191T→G transversion is a mutation in the *MT-ND3* gene encoding NADH dehydrogenase subunit 1, giving rise to a wide range of clinical diseases such as Leigh syndrome. The implementation of PGT in a woman carrying the m.10191T→G mutation with an affected child harboring 78% heteroplasmy in muscle showed that none of the unfertilized oocytes, arrested cleaved embryos, and cleaved embryos inherited the mutation in two distinct cycles, indicating a low transmission rate of the m.10191T→G mutation. However, live birth was not achieved as a consequence of transferring two mutation-free euploid embryos. Considering the low probability of transmission, the patient underwent natural conception, which led to the birth of a healthy son, lacking any mutation in his chorionic villi (Spath et al. 2021).

Overall, these results suggest that the m.10191T→G mutation has a low transmission rate, questioning the applicability of PGT for it.

The m.3697G→a mutation

The m.3697G→A transition is a mutation in the *MT-ND1* gene encoding NADH dehydrogenase subunit 1, which gives rise to Leigh syndrome. The results of two PGT cycles in an asymptomatic carrier with 52.3% heteroplasmy in blood for the m.3697G→A mutation demonstrated a transmission rate of 100% with TE heteroplasmy ranging from 15.2% to 100%. There were high levels of compatibility among individual blastomeres of a given embryo in terms of mutation loads (0–10.3% interblastomere variation). Transfer of a euploid blastocyst harboring 53.9% heteroplasmy resulted in a clinical pregnancy, which was terminated owing to the presence of anatomic defects in the fetus. Cells of amniotic fluid and various tissues of the aborted fetus were found to contain 49.7% and 47.7% mutation loads, respectively, indicating that the fetus's heteroplasmy could be represented by that of TE cells. In addition, no significant correlation existed between the mutation load of TE and that of either cumulus cells or PBs (Ji et al. 2021).

Altogether, the m.3697G→A mutation seems to have great inheritance potential, but uniform distribution among the blastomeres of a given embryo, highlighting the utility of PGT for lowering the risk of transmitting high levels of heteroplasmy.

Limitations of PGT for mtDNA disorders

Although several live births were documented following IVF/PGT cycles in couples harboring mtDNA mutations, this strategy has its limitations. Quantifying mutation loads in blastomeres or TE cells of preimplantation embryos in PGT programs are commonly carried out by restriction enzyme-based semiquantitative PCR techniques, which do not present high sensitivity and thus may lead to the underestimation of heteroplasmy levels. As an obvious example, the reanalysis of blood and urine specimens from a child, who was born as a result of two IVF cycles and TE biopsy followed by the transfer of a blastocyst containing 12% heteroplasmy for m.3243A→G, by a more sophisticated, accurate, and precise modality, namely ARMS-qPCR, indicated mutation loads of 46% and 42% at the age of 18 months, respectively (Mitalipov et al. 2014). However, this limitation can be overcome by implementing sensitive and high-throughput techniques for determining heteroplasmy levels in embryonic biopsy samples. In this regard, advanced qPCR assays, including TaqMan approach, ARMS-qPCR, or droplet digital PCR (ddPCR), and particularly next-generation sequencing (NGS)-based modalities, including single-cell RNA sequencing (scRNA-Seq) and single-cell transposase-accessible chromatin with sequencing (scATAC-Seq) raise the possibility of detecting rare and low loads of mutation (Marshall and Jones 2021; Parakatselaki and Ladoukakis 2021), and thus can promote the efficacy of PGT programs. It is noteworthy that single-cell DNA sequencing (scDNA-Seq) may offer a more reliable diagnostic value compared to scRNA-Seq, as it raises the possibility of detecting mutations of tRNA genes that affect the biological stability of tRNA transcripts, and consequently, may be undetectable *via* scRNA-Seq. Combining the molecular findings of different NGS approaches can also present even superior detection performance (He et al. 2020).

Mutation-free embryos are scarcely ever produced in IVF/PGT cycles, and as a result, PGT is based on the selection of embryos with mutation loads lower than a specific threshold. This is important because any level of heteroplasmy that exists in the embryo can potentially give rise to the expression of mtDNA diseases immediately after birth or in a late-onset manner due to the probability of stochastic and rapid segregation of mutant mtDNA, i.e., mtDNA bottleneck, throughout the embryofetal and postnatal development (Mitalipov et al. 2014). From the mechanistic perspective, the propagation of mtDNA heteroplasmy occurs in a mutation- and tissue-dependent fashion. In mitotically inactive cells and tissues, i.e., skeletal muscle, heart, and brain, mutant mtDNA(s) can stochastically replicate through a process called relaxed replication, leading to the rise of mutation load even higher than the biochemical threshold that is tolerable for cells, which results in the loss of cell function and disease. Regarding cells and tissues with high rates of proliferation and turnover, respectively, mtDNA molecules are transmitted through vegetative segregation, a process in which wild-type and mutant mtDNAs are asymmetrically divided between daughter cells during cytokinesis, bringing about the emergence of cells with different levels of heteroplasmy. Mitochondrial dysfunction and resultant cell death due to the overwhelming accumulation of pathogenic mtDNA mutation can lead to complete deletion of the mutation in the next generations, the process that represents negative selection, also known as purifying selection (van den Ameele et al. 2020; Chiaratti and Chinnery 2022). Mitophagy, a process that can be regulated by Ca^{2+} (Zhang J et al. 2022) and specific drugs, e.g., artesunate (Zhang K et al. 2022), can play a role in the clearance of mutant mtDNA at the organelle level. On the other hand, a preferential propagation of a specific mtDNA mutation can also occur, a phenomenon called positive selection or selfish propagation. Although underlying mechanisms are yet to be unraveled in humans, superior survival under particular conditions, improved mitochondrial membrane potential without damaging oxidative phosphorylation, and superior replication because of differences in mtDNA non-coding sequence (D-loop) can be possible explanations for positive selection at the levels of organism, cell, and mtDNA genome, respectively (van den Ameele et al. 2020; Chiaratti and Chinnery 2022). Accordingly, PGT alone is not sufficient to prevent transmitting heteroplasmic mtDNA diseases, and children born as a consequence of IVF/PGT cycles should be subjected to genetic analysis and follow-up in terms of mtDNA heteroplasmy.

As outlined previously, the success of PGT entails developing embryos in which mutation loads are lower than the crucial upper limit for the manifestation of the disorder. However, there are no precise criteria for determining safe levels of heteroplasmy, as this probably relies on the type of mutation, the unique presentations of the disorder, and family history (Poulton and Bredenoord 2010). Depending on the type of mutation, human mtDNA segregation follows one of three mechanisms of random genetic drift, negative selection, or positive selection, indicating potential variability of threshold value with selective inheritance pattern of mtDNA mutations (Otten et al. 2018). Consistently, a recent study failed to determine an asymptomatic upper limit for the m.3243A→G mutation (Ji et al. 2023). Based on a model of mtDNA heteroplasmy inheritance, the presence of more than 5% mutation loads in an embryo substantially increases the recurrence probability of the disorder (Samuels et al. 2013). Therefore, the transfer of embryos harboring less than 5% heteroplasmy may be of interest. Nevertheless, embryos with this characteristic may not be produced in IVF/PGT cycles. Such circumstances are the case in women who only produce oocytes with extremely high mutation loads, as well as those who carry homoplasmic mtDNA mutations. Fundamentally, PGT cannot be a suitable and efficacious clinical approach for these individuals, and they should pursue other reproductive alternatives, particularly mitochondrial replacement therapy.

Aside from the incidence of mtDNA disorders after birth, mtDNA heteroplasmy may compromise reproductive outcomes. As sources of cellular energy provision, mitochondria are deemed to have substantial involvement in the survival and development of human embryos, as well as in the adaption to bioenergetic demands, and consequently, maintenance of pregnancy. It was evident in mice that heteroplasmic mtDNA variants distinctly alter embryo metabolism, cellular competence, and induced pluripotent stem cell formation (Latorre-Pellicer et al. 2019). In humans, pathogenic mtDNA mutations were also shown to change the gene expression profile of preimplantation human embryos, particularly the genes that are vital for embryonic genome activation, and thereby impair embryo viability, cell differentiation, and development (Chatzovoulou et al. 2023), which in turn can ultimately reduce the ability of the embryo to establish successful implantation, pregnancy, and

live birth. In keeping with this hypothesis, numerous studies revealed that in women harboring mtDNA mutations, spontaneous pregnancy loss increases, and thus live birth decreases (Azadi et al. 2018; Feeney et al. 2019; Kuleva et al. 2019). However, a wide range of factors such as specific polymorphic variations other than the intervention on the blastocyst might contribute to spontaneous miscarriage (Tang et al. 2017).

Embryo biopsy is necessary in order to obtain the genetic material needed for PGT. Nowadays the most frequently utilized techniques for embryo biopsy in clinical routine are blastomere biopsy at the cleavage stage and TE biopsy at the blastocyst stage. Several negative outcomes, including abnormal epigenetic alterations (Wu et al. 2014), neurodegenerative diseases (Yu et al. 2009), and ovarian dysfunction (Yu et al. 2013) were reported to arise from blastomere biopsy in animal models. Although early evidence did not show negative effects (Cieslak-Janzen et al. 2006; Goossens et al. 2008), a robust study found that cleavage-stage embryo biopsy compromised subsequent embryo development and implantation competence in humans (Scott et al. 2013), possibly due to the relative fragility of the cleavage-stage embryo and particularly the elicitation of a substantial portion from embryo irrespective of its developmental fate (Cimadomo et al. 2016). Furthermore, a recent systematic review and meta-analysis indicated a positive association between blastomere biopsy at the cleavage stage and the likelihood of low birth weight and small for gestational age in newborns (Alteri et al. 2023). In comparison with cleavage-stage embryo biopsy, TE biopsy offers a more accurate and reliable diagnosis since it yields a higher amount of sample required for genetic testing. In addition, due to the elicitation of a scant and non-embryonic portion of the blastocyst, TE biopsy is less likely to jeopardize the developmental competence of the embryo (Cimadomo et al. 2016). Likewise, studies demonstrated that TE biopsy not only resulted in better blastocyst, implantation, and cumulative live birth outcomes compared to blastomere biopsy (Scott et al. 2013; Coll et al. 2018) but also did not bring about maternal and perinatal complications, including preterm delivery, gestational hypertension and diabetes mellitus, low birth weight, and birth defects (Mao et al. 2023), highlighting the advantage of transition from embryo biopsy at the cleavage-stage to TE biopsy in IVF clinics when PGT program is needed. However, there is also evidence of placental dysfunction during the future gestational period following TE biopsy (Ahlström et al. 2011; Zhang S et al. 2016). Moreover, the safety of TE biopsy in terms of obstetrical and neonatal outcomes is a matter of debate (Alteri et al. 2023). The centralization of labor services may at least reduce the probability of deleterious obstetrical outcomes and help to promote maternal health (Chen N and Pan 2022). Overall, regarding such uncertainties, recently-emerged noninvasive approaches based on analyzing DNA extracted from embryo spent medium may provide an opportunity to avoid the limitation of embryo biopsy (Ou et al. 2021), albeit the accuracy and precision of these strategies for the detection and measurement of mtDNA heteroplasmy remains to be determined.

Conclusion

PGT is a procedure that offers the opportunity to diminish the risk of transmitting genetic defects from parents to offspring by selecting and transferring embryos without genetic abnormalities. Given the great abundance of mtDNA copy numbers within cells, PGT is more applicable and accurate in the case of heteroplasmic mtDNA mutations, provided that a single blastomere can represent the whole embryo in terms of mutation load. Despite the presence of some contradictions, evidence on animal models supports the uniform distribution and stability of heteroplasmy among blastomeres of a given embryo during preimplantation development. With respect to humans, the inheritance properties of mtDNA mutations may differ depending on their type. The m.8993T→G mutation exhibits a skewed pattern of segregation, which gives rise to either the absence of mutation or the presence of high mutation loads in offspring, whereas the m.3243A→G, m.14487T→C, and m.3697G→A mutations display a high transmission rate, affecting almost all offspring with varied levels of heteroplasmy. Mutation load of TE in particular but also cleavage-stage blastomeres represent that of the entire embryo, as well as fetal and neonatal heteroplasmy for the m.8993T→G, m.3243A→G, m.14487T→C, and m.3697G→A mutations, corroborating the potential applicability of PGT for preventing the inheritance of these mutations. The m.10191T→G mutation seems to have low inheritance potential, consequently, PGT may be of low diagnostic value in the case of this mutation (Table 1).

Although available evidence is mostly in favor of the utility of PGT for common mtDNA mutations, this should be interpreted with caution. Firstly, not only the number of studies that investigated the

efficacy of PGT for mtDNA mutations is small but also sample sizes of these studies were small and often limited to one person. Secondly, PGT is faced with numerous problems and concerns, including low detection limits of assays quantifying mutant mtDNA, the probability of developing disease in embryos with mutation loads below the threshold, and the attenuation of embryo developmental competence as a result of cell removal, which restricts its applicability. Therefore, additional research with a long-term follow-up period for children born from IVF/PGT is warranted to confirm the advantage of PGT for preventing the transmission of mtDNA mutation.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Authors' contributions

Wrote original draft: XL, WLA; conceptualization, writing, review, and editing: JL.

References

Ahlström A, Westin C, Reismer E, Wikland M, Hardarson T. 2011. Trophectoderm morphology: an important parameter for predicting live birth after single blastocyst transfer. Hum Reprod. 26(12):3289–3296. doi:10.1093/humrep/der325.

Alteri A, Cermisoni GC, Pozzoni M, Gaeta G, Cavoretto PI, Viganò P. 2023. Obstetric, neonatal, and child health outcomes following embryo biopsy for preimplantation genetic testing. Hum Reprod Update. 29(3):291–306. doi:10.1093/humupd/dmad001.

Azadi A, Seo DJ, Jafari Sasansara H, Van Haute M. 2018. Mitochondrial DNA variations are associated with recurrent pregnancy loss. Mitochondrial DNA A DNA Mapp Seq Anal. 29(5):674–678. doi:10.1080/24701394.2017.1350854.

Bakare AB, Lesnefsky EJ, Iyer S. 2021. Leigh syndrome: a tale of two genomes. Front Physiol. 12:693734. doi:10.3389/fphys.2021.693734.

Calvo SE, Clauser KR, Mootha VK. 2016. MitoCarta2.0: an updated inventory of mammalian mitochondrial proteins. Nucleic Acids Res. 44(D1):D1251–1257. doi:10.1093/nar/gkv1003.

Chatzovoulou K, Mayeur A, Cagnard N, Zarhrate M, Bole C, Nitschke P, Jabot-Hanin F, Rötig A, Monnot S, Munnich A, et al. 2023. A shared pattern of altered gene expression in human embryos affected by mitochondrial diseases. Hum Reprod. 38(5):992–1002. doi:10.1093/humrep/dead052.

Chen N, Pan J. 2022. The causal effect of delivery volume on severe maternal morbidity: an instrumental variable analysis in Sichuan, China. BMJ Glob Health. 7(5):e008428. doi:10.1136/bmjgh-2022-008428.

Chen Y, Chen L, Zhou Q. 2021. Genetic association between eNOS gene polymorphisms and risk of carotid atherosclerosis: a meta-analysis. Herz. 46(Suppl 2):253–264. doi:10.1007/s00059-020-04995-z.

Chiaratti MR, Chinnery PF. 2022. Modulating mitochondrial DNA mutations: factors shaping heteroplasmy in the germ line and somatic cells. Pharmacol Res. 185:106466. doi:10.1016/j.phrs.2022.106466.

Childs AM, Hutchin T, Pysden K, Highet L, Bamford J, Livingston J, Crow YJ. 2007. Variable phenotype including Leigh syndrome with a 9185T > C mutation in the MTATP6 gene. Neuropediatrics. 38(6):313–316. doi:10.1055/s-2008-1065355.

Cieslak-Janzen J, Tur-Kaspa I, Ilkevitch Y, Bernal A, Morris R, Verlinsky Y. 2006. Multiple micromanipulations for preimplantation genetic diagnosis do not affect embryo development to the blastocyst stage. Fertil Steril. 85(6):1826–1829. doi:10.1016/j.fertnstert.2005.11.059.

Cimadomo D, Capalbo A, Ubaldi FM, Scarica C, Palagiano A, Canipari R, Rienzi L. 2016. The impact of biopsy on human embryo developmental potential during preimplantation genetic diagnosis. Biomed Res Int. 2016:7193075–10. doi:10.1155/2016/7193075.

Coll L, Parriego M, Boada M, Devesa M, Arroyo G, Rodríguez I, Coroleu B, Vidal F, Veiga A. 2018. Transition from blastomere to trophectoderm biopsy: comparing two preimplantation genetic testing for aneuploidies strategies. Zygote. 26(3):191–198. doi:10.1017/S0967199418000084.

Craven L, Tang MX, Gorman GS, De Sutter P, Heindryckx B. 2017. Novel reproductive technologies to prevent mitochondrial disease. Hum Reprod Update. 23(5):501–519. doi:10.1093/humupd/dmx018.

Cree LM, Samuels DC, de Sousa Lopes SC, Rajasimha HK, Wonnapinij P, Mann JR, Dahl HH, Chinnery PF. 2008. A reduction of mitochondrial DNA molecules during embryogenesis explains the rapid segregation of genotypes. Nat Genet. 40(2):249–254. doi:10.1038/ng.2007.63.

Dahl HH, Thorburn DR, White SL. 2000. Towards reliable prenatal diagnosis of mtDNA point mutations: studies of nt8993 mutations in oocytes, fetal tissues, children and adults. Hum Reprod. 15(Suppl 2):246–255. doi:10.1093/humrep/15.suppl_2.246.

Dean NL, Battersby BJ, Ao A, Gosden RG, Tan SL, Shoubridge EA, Molnar MJ. 2003. Prospect of preimplantation genetic diagnosis for heritable mitochondrial DNA diseases. Mol Hum Reprod. 9(10):631–638. doi:10.1093/molehr/gag077.

Feeney CL, Lim AZ, Fagan E, Blain A, Bright A, Maddison J, Devine H, Stewart J, Taylor RW, Gorman GS, et al. 2019. A case-comparison study of pregnant women with mitochondrial disease – What to expect? BJOG. 126(11):1380–1389. doi:10.1111/1471-0528.15667.

Finsterer J. 2011. Inherited mitochondrial neuropathies. J Neurol Sci. 304(1–2):9–16. doi:10.1016/j.jns.2011.02.012.

Goossens V, De Rycke M, De Vos A, Staessen C, Michiels A, Verpoest W, Van Steirteghem A, Bertrand C, Liebaers I, Devroey P, et al. 2008. Diagnostic efficiency, embryonic development and clinical outcome after the biopsy of one or two blastomeres for preimplantation genetic diagnosis. Hum Reprod. 23(3):481–492. doi:10.1093/humrep/dem327.

Gorman GS, Chinnery PF, DiMauro S, Hirano M, Koga Y, McFarland R, Suomalainen A, Thorburn DR, Zeviani M, Turnbull DM. 2016. Mitochondrial diseases. Nat Rev Dis Primers. 2(1):16080. doi:10.1038/nrdp.2016.80.

Gorman GS, Schaefer AM, Ng Y, Gomez N, Blakely EL, Alston CL, Feeney C, Horvath R, Yu-Wai-Man P, Chinnery PF, et al. 2015. Prevalence of nuclear and mitochondrial DNA mutations related to adult mitochondrial disease. Ann Neurol. 77(5):753–759. doi:10.1002/ana.24362.

Handyside AH, Kontogianni EH, Hardy K, Winston RM. 1990. Pregnancies from biopsied human preimplantation embryos sexed by Y-specific DNA amplification. Nature. 344(6268):768–770. doi:10.1038/344768a0.

He B, Dai C, Lang J, Bing P, Tian G, Wang B, Yang J. 2020. A machine learning framework to trace tumor tissue-of-origin of 13 types of cancer based on DNA somatic mutation. Biochim Biophys Acta Mol Basis Dis. 1866(11):165916. doi:10.1016/j.bbadis.2020.165916.

Heindryckx B, Neupane J, Vandewoestyne M, Christodoulou C, Jackers Y, Gerris J, Van den Abbeel E, Van Coster R, Deforce D, De Sutter P. 2014. Mutation-free baby born from a mitochondrial encephalopathy, lactic acidosis and stroke-like syndrome carrier after blastocyst trophectoderm preimplantation genetic diagnosis. Mitochondrion. 18:12–17. doi:10.1016/j.mito.2014.08.005.

Hellebrekers DM, Wolfe R, Hendrickx AT, de Coo IF, de Die CE, Geraedts JP, Chinnery PF, Smeets HJ. 2012. PGD and heteroplasmic mitochondrial DNA point mutations: a systematic review estimating the chance of healthy offspring. Hum Reprod Update. 18(4):341–349. doi:10.1093/humupd/dms008.

Holt IJ, Harding AE, Petty RK, Morgan-Hughes JA. 1990. A new mitochondrial disease associated with mitochondrial DNA heteroplasmy. Am J Hum Genet. 46(3):428–433.

Ji D, Li X, Pan J, Zong K, Chen D, Marley JL, Zou W, Deng X, Cao Y, Zhang Z, et al. 2021. Preimplantation genetic diagnosis for a carrier with m.3697G > A mitochondrial DNA mutation. J Assist Reprod Genet. 38(12):3251–3260. doi:10.1007/s10815-021-02354-3.

Ji D, Zhang N, Zou W, Zhang Z, Marley JL, Liu Z, Liang C, Shen L, Liu Y, Liang D, et al. 2023. Modeling-based prediction tools for preimplantation genetic testing of mitochondrial DNA diseases: estimating symptomatic thresholds, risk, and chance of success. J Assist Reprod Genet. 40(9):2185–2196. doi:10.1007/s10815-023-02880-2.

Kann O, Kovács R. 2007. Mitochondria and neuronal activity. Am J Physiol Cell Physiol. 292(2):C641–657. doi:10.1152/ajpcell.00222.2006.

Kuleva M, Ben Miled S, Steffann J, Bonnefont JP, Rondeau S, Ville Y, Munnich A, Salomon LJ. 2019. Increased incidence of obstetric complications in women carrying mitochondrial DNA mutations: a retrospective cohort study in a single tertiary centre. BJOG. 126(11):1372–1379. doi:10.1111/1471-0528.15515.

Lake NJ, Compton AG, Rahman S, Thorburn DR. 2016. Leigh syndrome: one disorder, more than 75 monogenic causes. Ann Neurol. 79(2):190–203. doi:10.1002/ana.24551.

Latorre-Pellicer A, Lechuga-Vieco AV, Johnston IG, Hämäläinen RH, Pellico J, Justo-Méndez R, Fernández-Toro JM, Clavería C, Guaras A, Sierra R, et al. 2019. Regulation of mother-to-offspring transmission of mtDNA Heteroplasmy. Cell Metab. 30(6):1120–1130.e1125. doi:10.1016/j.cmet.2019.09.007.

Lee HS, Ma H, Juanes RC, Tachibana M, Sparman M, Woodward J, Ramsey C, Xu J, Kang EJ, Amato P, et al. 2012. Rapid mitochondrial DNA segregation in primate preimplantation embryos precedes somatic and germline bottleneck. Cell Rep. 1(5):506–515. doi:10.1016/j.celrep.2012.03.011.

Mao D, Xu J, Sun L. 2023. Impact of trophectoderm biopsy for preimplantation genetic testing on obstetric and neonatal outcomes: a meta-analysis. Am J Obstet Gynecol. 2023:10. doi:10.1016/j.ajog.2023.08.010.

Marshall AS, Jones NS. 2021. Discovering cellular Mitochondrial Heteroplasmy heterogeneity with single cell RNA and ATAC sequencing. Biology. 10(6):503. doi:10.3390/biology10060503.

McFarland R, Taylor RW, Turnbull DM. 2010. A neurological perspective on mitochondrial disease. Lancet Neurol. 9(8):829–840. doi:10.1016/S1474-4422(10)70116-2.

Mitalipov S, Amato P, Parry S, Falk MJ. 2014. Limitations of preimplantation genetic diagnosis for mitochondrial DNA diseases. Cell Rep. 7(4):935–937. doi:10.1016/j.celrep.2014.05.004.

Monnot S, Gigarel N, Samuels DC, Burlet P, Hesters L, Frydman N, Frydman R, Kerbrat V, Funalot B, Martinovic J, et al. 2011. Segregation of mtDNA throughout human embryofetal development: m.3243A > G as a model system. Hum Mutat. 32(1):116–125. doi:10.1002/humu.21417.

Neupane J, Ghimire S, Vandewoestyne M, Lu Y, Gerris J, Van Coster R, Deroo T, Deforce D, Vansteelandt S, De Sutter P, et al. 2015. Cellular heterogeneity in the level of mtDNA heteroplasmy in mouse embryonic stem cells. Cell Rep. 13(7):1304–1309. doi:10.1016/j.celrep.2015.10.019.

Neupane J, Vandewoestyne M, Heindryckx B, Ghimire S, Lu Y, Qian C, Lierman S, Van Coster R, Gerris J, Deroo T, et al. 2014. A systematic analysis of the suitability of preimplantation genetic diagnosis for mitochondrial diseases in a heteroplasmic mitochondrial mouse model. Hum Reprod. 29(4):852–859. doi:10.1093/humrep/deu016.

Nightingale H, Pfeffer G, Bargiela D, Horvath R, Chinnery PF. 2016. Emerging therapies for mitochondrial disorders. Brain. 139(Pt 6):1633–1648. doi:10.1093/brain/aww081.

Otten ABC, Sallevelt S, Carling PJ, Dreesen J, Drüsedau M, Spierts S, Paulussen ADC, de Die-Smulders CEM, Herbert M, Chinnery PF, et al. 2018. Mutation-specific effects in germline transmission of pathogenic mtDNA variants. Hum Reprod. 33(7):1331–1341. doi:10.1093/humrep/dey114.

Ou Z, Deng Y, Liang Y, Chen Z, Sun L. 2021. Improved non-invasive preimplantation genetic testing for beta-thalassemia using spent embryo culture medium containing blastocoelic fluid. Front Endocrinol. 12:793821. doi:10.3389/fendo.2021.793821.

Parakatselaki ME, Ladoukakis ED. 2021. mtDNA Heteroplasmy: origin, detection, significance, and evolutionary consequences. Life. 11(7):633. doi:10.3390/life11070633.

Poulton J, Bredenoord AL. 2010. 174th ENMC international workshop: applying pre-implantation genetic diagnosis to mtDNA diseases: implications of scientific advances 19–21 March 2010, Naarden, The Netherlands. Neuromuscul Disord. 20(8):559–563. doi:10.1016/j.nmd.2010.05.008.

Poulton J, Macaulay V, Marchington DR. 1998. Mitochondrial genetics'98 is the bottleneck cracked? Am J Hum Genet. 62(4):752–757. doi:10.1086/301811.

Pyle A, Hudson G, Wilson IJ, Coxhead J, Smertenko T, Herbert M, Santibanez-Koref M, Chinnery PF. 2015. Extreme-depth re-sequencing of mitochondrial DNA finds no evidence of paternal transmission in humans. PLoS Genet. 11(5):e1005040. doi:10.1371/journal.pgen.1005040.

Reynier P, May-Panloup P, Chrétien MF, Morgan CJ, Jean M, Savagner F, Barrière P, Malthièry Y. 2001. Mitochondrial DNA content affects the fertilizability of human oocytes. Mol Hum Reprod. 7(5):425–429. doi:10.1093/molehr/7.5.425.

Ruhoy IS, Saneto RP. 2014. The genetics of Leigh syndrome and its implications for clinical practice and risk management. Appl Clin Genet. 7:221–234. doi:10.2147/TACG.S46176.

Sallevelt S, Dreesen J, Coonen E, Paulussen ADC, Hellebrekers D, de Die-Smulders CEM, Smeets HJM, Lindsey P. 2017. Preimplantation genetic diagnosis for mitochondrial DNA mutations: analysis of one blastomere suffices. J Med Genet. 54(10):693–697. doi:10.1136/jmedgenet-2017-104633.

Sallevelt SC, Dreesen JC, Drüsedau M, Hellebrekers DM, Paulussen AD, Coonen E, van Golde RJ, Geraedts JP, Gianaroli L, Magli MC, et al. 2017. PGD for the m.14487 T > C mitochondrial DNA mutation resulted in the birth of a healthy boy. Hum Reprod. 32(3):698–703. doi:10.1093/humrep/dew356.

Sallevelt SC, Dreesen JC, Drüsedau M, Spierts S, Coonen E, van Tienen FH, van Golde RJ, de Coo IF, Geraedts JP, de Die-Smulders CE, et al. 2013. Preimplantation genetic diagnosis in mitochondrial DNA disorders: challenge and success. J Med Genet. 50(2):125–132. doi:10.1136/jmedgenet-2012-101172.

Samuels DC, Wonnapinij P, Chinnery PF. 2013. Preventing the transmission of pathogenic mitochondrial DNA mutations: can we achieve long-term benefits from germline gene transfer? Hum Reprod. 28(3):554–559. doi:10.1093/humrep/des439.

Santorelli FM, Tanji K, Shanske S, DiMauro S. 1997. Heterogeneous clinical presentation of the mtDNA NARP/T8993G mutation. Neurology. 49(1):270–273. doi:10.1212/wnl.49.1.270.

Scott RT, Jr., Upham KM, Forman EJ, Zhao T, Treff NR. 2013. Cleavage-stage biopsy significantly impairs human embryonic implantation potential while blastocyst biopsy does not: a randomized and paired clinical trial. Fertil Steril. 100(3):624–630. doi:10.1016/j.fertnstert.2013.04.039.

Smeets HJ, Sallevelt SC, Dreesen JC, de Die-Smulders CE, de Coo IF. 2015. Preventing the transmission of mitochondrial DNA disorders using prenatal or preimplantation genetic diagnosis. Ann NY Acad Sci. 1350(1):29–36. doi:10.1111/nyas.12866.

Song WH, Yi YJ, Sutovsky M, Meyers S, Sutovsky P. 2016. Autophagy and ubiquitin-proteasome system contribute to sperm mitophagy after mammalian fertilization. Proc Natl Acad Sci USA. 113(36):E5261–5270.

Spath K, Babariya D, Konstantinidis M, Lowndes J, Child T, Grifo JA, Poulton J, Wells D. 2021. Clinical application of sequencing-based methods for parallel preimplantation genetic testing for mitochondrial DNA disease and aneuploidy. Fertil Steril. 115(6):1521–1532. doi:10.1016/j.fertnstert.2021.01.026.

Steffann J, Frydman N, Gigarel N, Burlet P, Ray PF, Fanchin R, Feyereisen E, Kerbrat V, Tachdjian G, Bonnefont JP, et al. 2006. Analysis of mtDNA variant segregation during early human embryonic development: a tool for successful NARP preimplantation diagnosis. J Med Genet. 43(3):244–247. doi:10.1136/jmg.2005.032326.

Steffann J, Gigarel N, Corcos J, Bonnière M, Encha-Razavi F, Sinico M, Prevot S, Dumez Y, Yamgnane A, Frydman R, et al. 2007. Stability of the m.8993T->G mtDNA mutation load during human embryofetal development has implications for the feasibility of prenatal diagnosis in NARP syndrome. J Med Genet. 44(10):664–669. doi:10.1136/jmg.2006.048553.

Steffann J, Gigarel N, Samuels DC, Monnot S, Borghese R, Hesters L, Frydman N, Burlet P, Frydman R, Benachi A, et al. 2014. Data from artificial models of mitochondrial DNA disorders are not always applicable to humans. Cell Rep. 7(4):933–934. doi:10.1016/j.celrep.2014.05.005.

Steffann J, Monnot S, Magen M, Assouline Z, Gigarel N, Ville Y, Salomon L, Bessiere B, Martinovic J, Rötig A, et al. 2021. A retrospective study on the efficacy of prenatal diagnosis for pregnancies at risk of mitochondrial DNA disorders. Genet Med. 23(4):720–731. doi:10.1038/s41436-020-01043-3.

Stewart JB, Chinnery PF. 2015. The dynamics of mitochondrial DNA heteroplasmy: implications for human health and disease. Nat Rev Genet. 16(9):530–542. doi:10.1038/nrg3966.

Tajima H, Sueoka K, Moon SY, Nakabayashi A, Sakurai T, Murakoshi Y, Watanabe H, Iwata S, Hashiba T, Kato S, et al. 2007. The development of novel quantification assay for mitochondrial DNA heteroplasmy aimed at preimplantation genetic diagnosis of Leigh encephalopathy. J Assist Reprod Genet. 24(6):227–232. doi:10.1007/s10815-007-9114-0.

Tang L, Zheng S, Wang Y, Li F, Bao M, Zeng J, Xiang J, Luo H, Li J. 2017. Rs4265085 in GPER1 gene increases the risk for unexplained recurrent spontaneous abortion in Dai and Bai ethnic groups in China. Reprod Biomed Online. 34(4):399–405. doi:10.1016/j.rbmo.2017.01.005.

Tatuch Y, Christodoulou J, Feigenbaum A, Clarke JT, Wherret J, Smith C, Rudd N, Petrova-Benedict R, Robinson BH. 1992. Heteroplasmic mtDNA mutation (T−−G) at 8993 can cause Leigh disease when the percentage of abnormal mtDNA is high. Am J Hum Genet. 50(4):852–858.

Taylor RW, Turnbull DM. 2005. Mitochondrial DNA mutations in human disease. Nat Rev Genet. 6(5):389–402. doi:10.1038/nrg1606.

Treff NR, Campos J, Tao X, Levy B, Ferry KM, Scott RT. Jr. 2012. Blastocyst preimplantation genetic diagnosis (PGD) of a mitochondrial DNA disorder. Fertil Steril. 98(5):1236–1240. doi:10.1016/j.fertnstert.2012.07.1119.

van den Ameele J, Li AYZ, Ma H, Chinnery PF. 2020. Mitochondrial heteroplasmy beyond the oocyte bottleneck. Semin Cell Dev Biol. 97:156–166. doi:10.1016/j.semcdb.2019.10.001.

Wai T, Teoli D, Shoubridge EA. 2008. The mitochondrial DNA genetic bottleneck results from replication of a subpopulation of genomes. Nat Genet. 40(12):1484–1488. doi:10.1038/ng.258.

Wang K, Xu Y, Sun Q, Long J, Liu J, Ding J. 2018. Mitochondria regulate cardiac contraction through ATP-dependent and independent mechanisms. Free Radic Res. 52(11–12):1256–1265. doi:10.1080/10715762.2018.1453137.

White SL, Collins VR, Wolfe R, Cleary MA, Shanske S, DiMauro S, Dahl HH, Thorburn DR. 1999. Genetic counseling and prenatal diagnosis for the mitochondrial DNA mutations at nucleotide 8993. Am J Hum Genet. 65(2):474–482. doi:10.1086/302488.

Wu H, Whitcomb BW, Huffman A, Brandon N, Labrie S, Tougias E, Lynch K, Rahil T, Sites CK, Pilsner JR. 2019. Associations of sperm mitochondrial DNA copy number and deletion rate with fertilization and embryo development in a clinical setting. Hum Reprod. 34(1):163–170. doi:10.1093/humrep/dey330.

Wu Y, Lv Z, Yang Y, Dong G, Yu Y, Cui Y, Tong M, Wang L, Zhou Z, Zhu H, et al. 2014. Blastomere biopsy influences epigenetic reprogramming during early embryo development, which impacts neural development and function in resulting mice. Cell Mol Life Sci. 71(9):1761–1774. doi:10.1007/s00018-013-1466-2.

Xiang Q, Cao Y, Xu H, Yang Z, Tang L, Xiang J, Li J, Deng H, Yuan L. 2021. Novel MFSD8 variants in a chinese family with nonsyndromic macular dystrophy. J Ophthalmol. 2021:6684045–6684045. doi:10.1155/2021/6684045.

Yu Y, Wu J, Fan Y, Lv Z, Guo X, Zhao C, Zhou R, Zhang Z, Wang F, Xiao M, et al. 2009. Evaluation of blastomere biopsy using a mouse model indicates the potential high risk of neurodegenerative disorders in the offspring. Mol Cell Proteomics. 8(7):1490–1500. doi:10.1074/mcp.M800273-MCP200.

Yu Y, Zhao Y, Li R, Li L, Zhao H, Li M, Sha J, Zhou Q, Qiao J. 2013. Assessment of the risk of blastomere biopsy during preimplantation genetic diagnosis in a mouse model: reducing female ovary function with an increase in age by proteomics method. J Proteome Res. 12(12):5475–5486. doi:10.1021/pr400366j.

Zhang H, Burr SP, Chinnery PF. 2018. The mitochondrial DNA genetic bottleneck: inheritance and beyond. Essays Biochem. 62(3):225–234. doi:10.1042/EBC20170096.

Zhang J, Shen Q, Ma Y, Liu L, Jia W, Chen L, Xie J. 2022. Calcium Homeostasis in Parkinson's Disease: from pathology to treatment. Neurosci Bull. 38(10):1267–1270. doi:10.1007/s12264-022-00899-6.

Zhang K, Yang Y, Ge H, Wang J, Lei X, Chen X, Wan F, Feng H, Tan L. 2022. Neurogenesis and proliferation of neural stem/progenitor cells conferred by artesunate via FOXO3a/p27Kip1 Axis in Mouse Stroke Model. Mol Neurobiol. 59(8):4718–4729. doi:10.1007/s12035-021-02710-5.

Zhang S, Luo K, Cheng D, Tan Y, Lu C, He H, Gu Y, Lu G, Gong F, Lin G. 2016. Number of biopsied trophectoderm cells is likely to affect the implantation potential of blastocysts with poor trophectoderm quality. Fertil Steril. 105(5):1222–1227.e1224. doi:10.1016/j.fertnstert.2016.01.011.

Zhou Q, Li H, Li H, Nakagawa A, Lin JL, Lee ES, Harry BL, Skeen-Gaar RR, Suehiro Y, William D, et al. 2016. Mitochondrial endonuclease G mediates breakdown of paternal mitochondria upon fertilization. Science. 353(6297):394–399. doi:10.1126/science.aaf4777.

Zhu Y, Huang R, Wu Z, Song S, Cheng L, Zhu R. 2021. Deep learning-based predictive identification of neural stem cell differentiation. Nat Commun. 12(1):2614. doi:10.1038/s41467-021-22758-0.